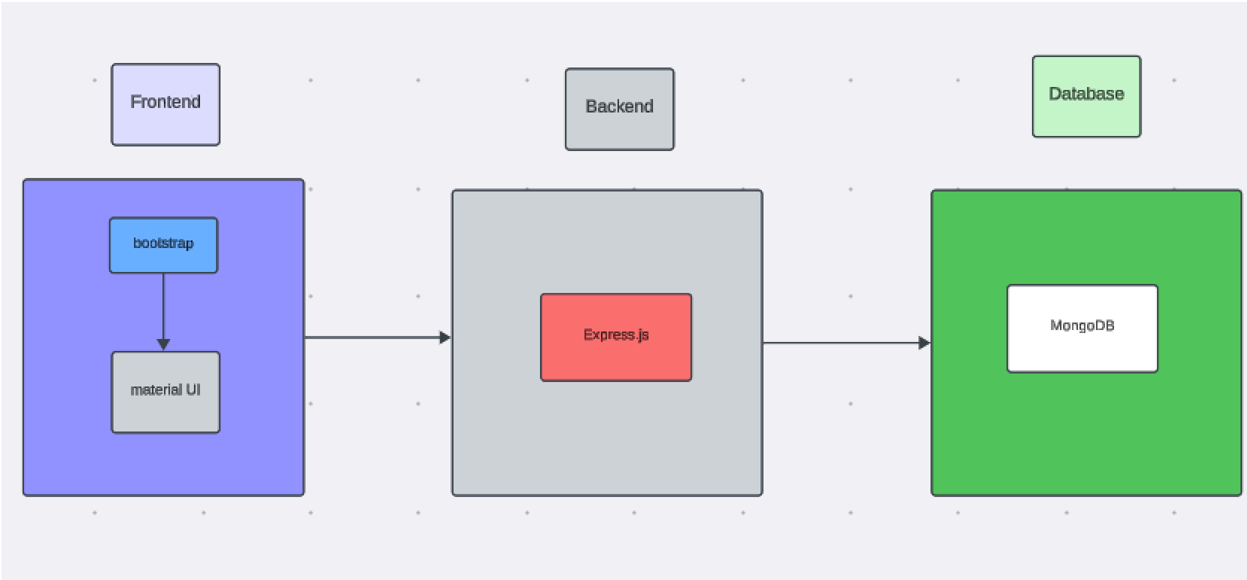

Technical Architecture of the Online Complaint Registration and Management System

The technical architecture of our **Online Complaint Registration and Management System** follows a **client-server architecture**, where the **frontend** acts as the client and the **backend** functions as the server. The frontend is responsible for providing an interactive and user-friendly interface for customers, agents, and administrators. It communicates with the backend through **RESTful APIs** using the **Axios** library, enabling efficient data exchange and seamless interaction between the client and server.

The frontend is developed using **React.js** and utilizes **Bootstrap** and **Material UI** to create a responsive, modern, and intuitive user interface. These libraries ensure that users can easily register, log in, submit complaints, track complaint status, manage profiles, and access dashboards across different devices.

On the backend, the application is built with **Node.js** and the **Express.js** framework, which manages server-side logic, authentication, complaint processing, user management, and API requests. Express.js provides a scalable and efficient environment for handling communication between the frontend and the database.

For data storage and retrieval, the system uses **MongoDB**, a NoSQL database that efficiently stores and manages information such as user accounts, complaint details, complaint categories, complaint status, assigned agents, and administrative records.

MongoDB ensures fast, reliable, and scalable access to data, enabling smooth complaint registration, tracking, and resolution.

The system also incorporates **JSON Web Tokens (JWT)** for secure user authentication and authorization, ensuring that only authorized users can access protected resources based on their roles, such as Customer, Agent, or Administrator.

Together, **React.js**, **Bootstrap**, **Material UI**, **Axios**, **Node.js**, **Express.js**, **MongoDB**, and **JWT** form a robust and scalable technical architecture for the Online Complaint Registration and Management System. This architecture enables secure user authentication, efficient complaint management, real-time status updates through API communication, reliable data storage, and a seamless user experience for customers, agents, and administrators.